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Research Assignment 1

Section A:

1. An organised collection of structured data. Databases are important as they are built to store large amounts of data and allow users to query & update information at the same time. Databases are used in e-commerce to store inventory & customer personal information. They are also used in banking to track balances & recording transactions.
2. A datawarehouse ~~store~~ combines data from multiple sources into a central location. Unlike a database it often contains current & historic data that has been extracted, transformed & loaded. It is often used in business as it allows data analysts to develop insights & guide business decisions.
3. It is a centralized repository which stores raw data (structured & unstructured). This is different from a datawarehouse which stores "clean" data. It is often used for machine learning & predictive analysis by data scientists.

4. A storage system that stores data for a specific business area/function. A data mart is a subset of an existing data warehouse. It is used by specific business departments eg. marketing & sales etc.
5. Structured data - Highly organised information stored in rows & columns in a structured table.
Eg. Spreadsheets

Semi Structured data - A combination of organized & unorganized information
Eg. Configuration files

Unstructured data - Raw, qualitative information that cannot be stored in a regular structured database.
Eg. Emails

6. Creating a visual mapping of data to show it will be collected, stored & used. It helps bridge business needs with technical implementation, which ensures consistent data quality while analysing. Conceptual data model defines the structure of business & data. Logical data model builds on the conceptual data model by including technical information.

7. The use of machine learning & statistics to discover patterns or anomalies in a large data set. It allows businesses to predict trends which inform business decisions.
- Eg. 1. Analysing purchase history to create target marketing campaigns.
- Eg. 2. Scanning millions of credit cards to detect fraudulent activities.

8. A software which allows users to create & manage databases.

MySQL - Widely used, open-source relational DBMS. Often used by e-commerce websites as it can handle high volume transactions smoothly.

PostgreSQL - Advanced, open source enterprise-class object relational database which supports both SQL & JSON querying. It is used by financial applications robust data integrity features.

Oracle - A proprietary, highly scalable multi-model relational database. It is used by massive corporations which require strong security & the ability to process millions of ~~queries~~ transactions across global servers simultaneously.

MongoDB - Open Source No Sql database which stores data in flexible JSON-like documents rather than tables. Used in applications which require handling unstructured data.

9. Process used to collect data, clean it & format it & store it in a data warehouse.

Extract - pull raw data from multiple sources i.e. APIs & SaaS platforms.

Transform - Improve data integrity by cleaning duplicates, handling missing values, standardizing formats

Load - Write the final structured data into a data warehouse for analysis.

It is important as it provides clean, structured data which can be used for business intelligence.

10. CSV - Used for small datasets (< 1000000 rows)

JSON - used for web APIs & log data with records that contain varying schema

Parquet - used for big data analytics

Avro - used for high volume of ~~real data~~ real time data streaming where schemas change overtime.

Section B :

11. A technology which allows computers to mimic human intelligence. Narrow AI is designed to perform tasks within a specific scope. Eg. Virtual assistants. General AI is a theoretical form of AI which mimics human intelligence & is able to solve problems & learn new skills similar to how humans do.
12. A subfield of AI which allows computers to learn using ~~at~~ algorithms in order to recognise patterns. Supervised learning involves training models using labelled datasets which allows them to improve in accuracy overtime. Unsupervised machine learning allows models to look for patterns in unlabelled data. Reinforcement learning uses trial & error to teach models by establishing a reward system.
13. A subfield of AI & ML which uses multilayered neural networks in order to solve complex problems & process large datasets. Deep learning is a specialized branch of ML which is able to process unstructured data & clean data with minimal human intervention. Neural networks are layers of interconnected nodes inspired by the human brain which perform pattern recognition & predictions without explicitly programming involved. Information is inputted in the input layer where input nodes process it & pass it on to the next layer. In the hidden layers, information is processed further & further layer to layer. The ~~pa~~ output layer provides the final results of processing which occurred in the AI neural network.

14. A subfield of computer science & AI that uses ML to allow computers to understand & also communicate human language. It is often used in generative AI (ie ChatGPT), voice assistants & translation apps
15. Deep learning models which are trained on a large amount of data in order to understand, generate & interpret human language. LLMs process unstructured data types which contains words from books, articles & internet forums. Examples of LLMs include ~~Gemini~~ Google Gemini & Anthropic's Claude which are capable of processing text, images, videos & code in order to return the output required by the user
16. Generative AI is a subset of AI which its recognition of patterns in a massive dataset in order to generate unique / original contents such as text, images, audio etc. It first involves training by finding patterns in raw data & practising with labelled examples that contain the correct answer. Next, it constantly improves by making a prediction by measuring how wrong it was & "fixing" its self to make a better guess next time. In order to read & write, the AI breaks down sentences into tokens, linking them to understand word meanings & track entire conversations.

Eg. Suno - Used to make music & instrumentals from prompts
Sora - Used to turn images into immersive videos
Cursor - Used to help developers build, debug & write code using natural language

17. A dataset which is used to teach ML models to identify patterns, make decisions & generate content. The quality of ~~training~~ training data is important as it improves the accuracy of the model ensuring ~~that~~ it outputs the required output by the user. In the event that training data is biased or incorrect, the ML model will hallucinate.

18. A subset of AI where models are trained using images & videos in order to understand the visual world. Raw data (images & videos) are fed to AI models that use ~~algorithm~~ algorithms to detect patterns between a large dataset of known patterns. This type of AI translates ~~photo~~ media into grid of number resembling pixels. It uses probability & prediction to calculate the probability of an object & assign it to a label. Real world uses include assisting doctors to automatically analyse X-rays, MRIs & CT scans and also enhance surveillance with motion detection & automated license plate reader

SECTION C

19. A set of protocols & rules that allow the communication between softwares. It is similar to a waiter in a restaurant. The customer (Your App) wants to see a lot of available food. The waiter (API) takes the request to the kitchen (The server) where it is prepared and handed back to the waiter. The waiter brings back the data back to the app. They are important as improve security by only sharing specific data & keeping the rest of the system hidden, improve efficiency as developers don't have to build features from scratch & also allows tools to sync data automatically in the background



Figure 1: A diagram showing how APIs work

20. A set of randomly generated numbers used to authenticate clients/users & grant access to the API. It facilitates secure communication between software systems.

For production applications, dedicated secret management systems are used to store API keys & for development apps, a .dev file is used to store them. If API keys are leaked to the public will grant users unauthorized access leading to unauthorized use, exposure of data, service abuse etc.

21. Representational State transfer API is an API that follows the principles of the REST architectural style.

GET - Retrieve data

POST - Create new data

PUT - Update existing data

DELETE - Remove data

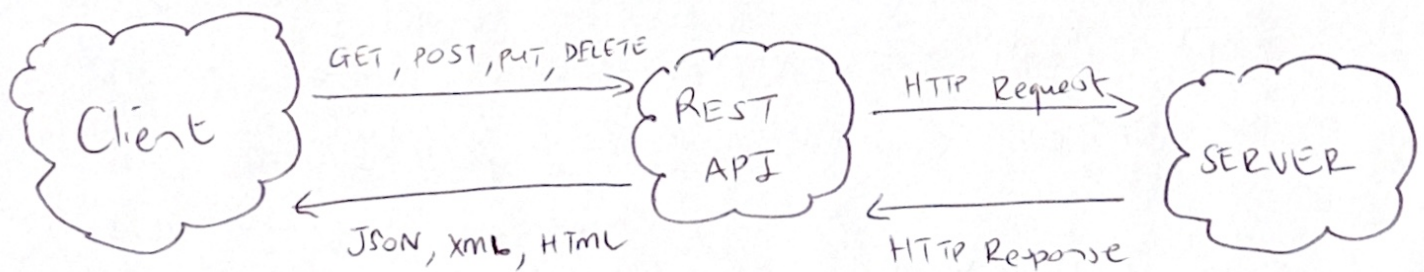


Figure 2: A diagram showing a rest API model

22. A lightweight data format used to transmit & store structured data. It is commonly used as it is text based which makes it easy to ^{be} generated & parsed by any programming language.

Example of JSON object:

```
{  
  "name": "Fuze",  
  "age": "26",  
  "employed": true,  
  "skills": ["Python", "SQL", "Databricks"]  
}
```

23. A way to construct a prompt to guide an AI model to produce a specific & accurate result. Prompt engineering is important as it ensures the AI understand what is required & produce meaningful response and also to minimize extensive postprocessing.

Poor Prompt: "Write a marketing email for my new app"

Correction: "Act as a growth marketing expert & write a launch email for "BrightFocus", a desktop app that blocks out social media distractions for remote workers. Keep it under 150 words. Use a casual tone. Include 3 benefits in bulleted list & a Call to Action link at the end."

Poor prompt: "Summarize this article"

Correction: "Summarize this attached article about climate change in London for executive meeting. Include a 3 sentence executive summary and also include points on financial impacts"

24. A map function is a higher order function used to apply a function on each item of a list or array. This replaces the need of using a ~~loop~~ manual loop.

Eg. `list Cmap(Str.upper, ['apple', 'banana'])`

⇓ output

`['APPLE', 'BANANA']`

25. The delivery of computing services over the internet.

IaaS (Infrastructure as a Service) - provides on demand access to computing resources (eg. servers, storage, networking) ~~databases & applications~~

PaaS (Platform as a Service) - provides an on demand environment for testing & deploying applications (Google App Engine)

Software as a Service (SaaS) - offers fully functional software applications over the internet (Microsoft 365)

26. A software which tracks & manages files overtime.

Git is a widely used distributed version control system which tracks changes, safely revert to past versions & collaborate without overwriting each others work. A commit is saving the state of your project at a specific point in time. Branch is used to move from the main line of development to build new features or fix bugs.

27. It is an open source web application that allows you to create documents containing live code, equations & visualization etc. Python is normally used in Jupyter notebooks. It is popular with data professionals as it breaks down projects into small, logical steps. This allows you to experiment, view data, debug specific sections & write documentation in one file.

28. A popular programming language which is widely used for web applications, ~~web dev~~ data science & machine learning. It is used in data analysis & AI for its simple syntax, versatility & a large ecosystem of specialized data libraries. It also works seamlessly with other technologies such as SQL databases & web APIs.

Pandas: Used for handling structured, tabular data using Dataframes

Numpy: Provides advanced high performance operations using multidimensional arrays.

Seaborn: Used to create graphs & charts

Beautiful Soap : Allows analysts to scrap data from the internet.

29. It describes how people's sensitive information is properly handled. Data privacy protects people from identity theft, ~~theft~~ fraud & manipulation by making sure people have control over how their personal information is collected, used & shared. GDPR is a governmental framework in the European Union which dictates how companies should handle & protect sensitive personal data.

30. Data Ethics provides guidance on how data should be handled.

Eg. 1. Cambridge Analytica illegally used personal information from Facebook's 87 million users for academic research

Eg. 2. Google used user location data with advertisers

Eg. 3. Twitter gave advertisers access to user data to improve marketing campaigns

31. A framework of policies that focus on the quality, security & availability of organizational data

Data Privacy: Ensuring data is consistent, accurate & standardized.

Data Stewardship: Encouraging people to enforce data related rules & resolve discrepancies.

Protection & Compliance: Securing accessibility, labelling sensitive information and following legal requirements

Change Management: Adopt AI technologies & undergo structural changes as the framework evolves

32. Bias results which are due to biasness in the ~~training~~^{training} of machine learning models. A good example would be automated CV screeners which unintentionally penalize female applicants by associating male ~~dominant~~ dominated phrasings with better success.

33.

	Role	Skills	Tools
Data Analyst	Extracts insights from data in order to help business make better decisions	Business intelligence, data visualization & statistical reporting	SQL, Python, PowerBI, Tableau, Excel
Data Scientist	Apply machine learning & statistical models to gain deeper insights from the data	Machine learning & statistics	Python, R, Jupyter notebook, SciKit learn
Data Engineer	Builds & maintains infrastructure that allows the flow of data in the organization	Data pipelines development, data warehousing, infrastructure management	Python, SQL, Apache Spark, AWS

34. Technological processes for collecting, managing & analysing business data in order to gain insight on its performance and inform business strategies. Power BI & Tableau are BI tools used to process large amounts of data & create quality data visualizations.

35. ~~Research~~ I found the research on Machine ^{Learning} ~~Learning~~ & AI specifically neural networks very interesting. I think it is profound how we look at biology in order to solve technological problems. The idea of mimicking how the human brain processes information and how it learns new topics and it all comes down to fundamentally the identification of patterns.

36. I found REST APIs rather difficult to grasp. I used Gemini AI to try & tone down the technical jargon & break down what the concept really is into small pieces of information. I also used Google Images in order to understand the architecture as I am a visual learner.

37. I think AI is here to be an auxiliary tool that will be used in order to execute tasks efficiently. I do not think it will replace the professional but will be an aid instead. Eg. Doctors will be able to diagnose cancers using AI which will boost the medical industry.

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